

Cyber Range Lab 2 - Firewall Hardening & Defense

Objective:

Students will isolate subnets using firewall policies and verify that targeted blocks do not affect overall node health.

Required Equipment:

DMZ Pi, Client Pi, Access to Server dashboard, and Terminal access

Steps:

1. Display Current Firewall Rules

```
sudo nft list ruleset
```

2. Insert Rule Blocking DMZ → Client

```
sudo nft add rule inet filter forward \  
ip saddr 10.0.0.20 ip daddr 10.0.0.40 drop
```

3. Verify Block

On DMZ Pi:

```
ping -c 1 10.0.0.40
```

Expected: failure.

4. Check Dashboard

Confirm that all nodes still show "Up" (block affects traffic, not health).

5. Award Points

```
curl -X POST http://10.0.0.30:8080/score/team1/15
```

Check Your Work:

Screenshot of nft rule insertion, failed ping example, dashboard unchanged, showing "Up," scoreboard update showing +15 points.

Reflection Question:

How does a targeted firewall block differ from disconnecting a node?